Instructions

- *This homework is* **not** *to be handed in or given a grade.*
- *There will be a* **quiz in class** *: Section B : 15/04/2026, Section A : 16/04/2026*
- *TA for this assignment is (theoretical part) :* ***Ali Saheb Pasand****.*

**Question 1.** Let $p(x, z)$ be the joint probability of a latent variable model where $x$ and $z$ denote the observed and unobserved variables, respectively. Let $q(z|x)$ be an auxiliary distribution which we call the *proposal*, and define [1]

$$\mathcal{L}_K[q(z|x)] = \int \cdots \int \left( q(z_1|x)...q(z_K|x) \log \frac{1}{K} \sum_{j=1}^{K} \frac{p(x, z_j)}{q(z_j|x)} \right) dz_1 dz_2 ... dz_K$$

We've seen in class that this objective is a tighter lower bound on $\log p(x)$ than the evidence lower bound (ELBO), which is equal to $\mathcal{L}_1$ ; that is $\mathcal{L}_1[q(z|x)] \leq \mathcal{L}_K[q(z|x)] \leq \log p(x)$.

In fact, $\mathcal{L}_K[q(z|x)]$ can be interpreted as the ELBO with a refined proposal distribution. For $z_j$ drawn i.i.d. from $q(z|x)$ with $2 \leq j \leq K$, define the *unnormalized* density

$$\tilde{q}(z|x, z_2, ..., z_K) := \frac{p(x, z)}{\frac{1}{K}\left(\frac{p(x,z)}{q(z|x)} + \sum_{j=2}^{K} \frac{p(x,z_j)}{q(z_j|x)}\right)}$$

*(Hint : in what follows, you might need to use the fact that if $w_1, ..., w_K$ are random variables that have the same distribution, then $K\mathbb{E}[w_1] = \sum_i \mathbb{E}[w_i] = \mathbb{E}[\sum_i w_i]$. You need to identify such $w_i$'s before applying this fact for each subquestion. )*

1. Show that $\mathcal{L}_K[q(z|x)] = \mathbb{E}_{z_{2:K}}[\mathcal{L}_1[\tilde{q}(z|x, z_2, ..., z_K)]]$ ; that is, the importance-weighted lower bound with $K$ samples is equal to the average ELBO with the unnormalized density as a refined proposal.
2. Show that $q_K(z|x) := \mathbb{E}_{z_{2:K}}[\tilde{q}(z|x, z_2, ..., z_K)]$ is in fact a probability density function. Also, show that $\mathcal{L}_1[q_K(z|x)]$ is an even tighter lower bound than $\mathcal{L}_K[q(z|x)]$. This implies $q_K(z|x)$ is closer to the true posterior $p(z|x)$ than $q(z|x)$ due to resampling, since $\mathcal{L}_K[q(z|x)] \geq \mathcal{L}_1[q(z|x)]$. (Hint : $f(x) := -x \log x$ is concave.)

1. Note that $\mathcal{L}_K[\cdot]$ is a "functional" whose input argument is a "function" $q(\cdot|x)$.

**Question 2.** In this task, we will go deeper into mathematics of diffusion models. Consider a denosing diffusion probabilistic model (DDPM) with the encoder process given by a linear Gaussian model : $q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) = \mathcal{N}(\boldsymbol{x}_t|\sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t I)$, where $\beta_t \in (0,1)$ is a fixed noise schedule. The forward diffusion process starts from initial image $\boldsymbol{x}_0$ from the dataset and ends at $\boldsymbol{x}_T$ ($T$ is a fixed number of steps). We assume $\boldsymbol{x}_T \sim p(\boldsymbol{x}_T) = \mathcal{N}(\boldsymbol{x}_T|0, I)$. The goal of the training is to learn a reversed (denoising process) $p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)$, which will allow to generate an image $\boldsymbol{x}_0$ starting from Gaussian noise $\boldsymbol{x}_T$.

2.1 Given the equation for linear Gauassian encoder process $q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1})$, show that the ground truth denoising process is

$$q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) = \mathcal{N}(\boldsymbol{x}_{t-1}|\tilde{\mu}_t(\boldsymbol{x}_t, \boldsymbol{x}_0), \tilde{\beta}_t I) \tag{1}$$

where

$$\tilde{\mu}_t(\boldsymbol{x}_t, \boldsymbol{x}_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}x_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t \tag{2}$$

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$$

$$\alpha_t = 1 - \beta_t$$

$$\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$$

If needed, you can use the following equation without proving it :

$$q(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}(\boldsymbol{x}_t|\sqrt{\bar{\alpha}_t}\boldsymbol{x}_0, (1-\bar{\alpha}_t)I) \tag{3}$$

Hint : use Bayes rule and Markovian property of the encoder process

2.2 As we saw in task 2.1, it is possible to reverse the diffusion process analytically, without training anything. Explain, why we still need to train the reverse process $p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)$ to generate images.

2.3 Now, let's derive the objective function for DDPM. Essentially, DDPM is a hierarchical variational autoencoder (with latent variables $\{x_1, \ldots, x_T\}$) and its objective is an evidence lower bound (ELBO) for $\log p(x_0)$. Show that

$$\log p(x_0) \geq \mathcal{L}_{DDPM}(\theta; \boldsymbol{x}_0) = -L_0(\boldsymbol{x}_0) - \sum_{t=2}^{T} L_{t-1}(\boldsymbol{x}_0) - L_T(\boldsymbol{x}_0)$$

where

- (reconstruction term) $L_0(\boldsymbol{x}_0) = -\mathbb{E}_{q(\boldsymbol{x}_1|\boldsymbol{x}_0)} \log p_\theta(\boldsymbol{x}_0|\boldsymbol{x}_1)$
- (denoising matching term) $L_{t-1}(\boldsymbol{x}_0) = \mathbb{E}_{q(\boldsymbol{x}_t|\boldsymbol{x}_0)} D_{\mathrm{KL}}(q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0)||p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t))$
- (prior matching term) $L_T(\boldsymbol{x}_0) = D_{\mathrm{KL}}(q(\boldsymbol{x}_T|\boldsymbol{x}_0)||p(\boldsymbol{x}_T))$

The following equations migth be useful for derivations :

$$p_\theta(\boldsymbol{x}_{0:T}) = p(\boldsymbol{x}_T) \prod_{t=1}^{T} p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)$$

$$q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) = q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}, \boldsymbol{x}_0) = \frac{q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0)q(\boldsymbol{x}_t|\boldsymbol{x}_0)}{q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_0)}$$

2.4 Which term in $\mathcal{L}_{DDPM}$ doesn't affect optimisation over parameters and therefore can be excluded from the objective function ?

2.5 Compare ELBO for vanila VAE and ELBO for DDPM. What is the key difference between them (in terms of trainable parameters) ?

2.6 Let's consider $L_{t-1}(\boldsymbol{x}_0)$ and $L_0(x_0)$ in more detail.

- Using eq. 2 and 3, show that $\tilde{\mu}_t(\boldsymbol{x}_t, \boldsymbol{x}_0) = \frac{1}{\sqrt{\alpha_t}}\big(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}\big)$, where $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\epsilon}|0, I)$
- A common parametrisation for denoising process is $p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) = \mathcal{N}(\boldsymbol{x}_{t-1}|\mu_\theta(\boldsymbol{x}_t, t), \sigma_t^2 I)$, where mean of the Gaussian $\mu_\theta(\boldsymbol{x}_t, t)$ is trainable (here we consider $\sigma_t^2$ to be fixed for simplicity, while in practice it is trainable). However, instead of training a model to predict the $\mu_\theta(\boldsymbol{x}_t, t)$ directly, a common choice is to train a neural network $\boldsymbol{\epsilon}_\theta$ (aka "denoiser") to predict only the noise term : $\mu_\theta(\boldsymbol{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}\big(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)\big)$. Show that

$$\mathbb{E}_{q(\boldsymbol{x}_0)} L_{t-1}(\boldsymbol{x}_0) = \mathbb{E}_{q(\boldsymbol{x}_0), \boldsymbol{\epsilon}\sim\mathcal{N}(0,I)}\Big[\lambda_t ||\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\big(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t\big)||^2\Big] + const \tag{4}$$

  where $\lambda_t = \frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)}$ and $q(\boldsymbol{x}_0)$ is the groundtruth data distribution
  Hint : you can use the equation for KL divergence between multivariate normal distributions without deriving it.
- Show that $\mathbb{E}_{q(\boldsymbol{x}_0)} L_0(\boldsymbol{x}_0)$ can be written in the same way as eq. 4

2.7 Finally, put together the equations for ELBO terms and get the DDPM loss function.

**Question 3.** This assignment visits two methods developed for solving stability and training issues in GAN formulation.

3.1 **GAN Stability via Extragradient** In this question, we analyze the training dynamics of GANs using a simple bilinear game. Consider the following value function :

$$V(d, g) = dg \tag{5}$$

with $g \in \mathbb{R}$ and $d \in \mathbb{R}$. We will use this simple example to study oscillations and stabilization techniques in GAN training.

(a) **Simultaneous Gradient Descent/Ascent** Consider gradient descent/ascent with learning rate $\alpha$ to iteratively minimize $V(d, g)$ w.r.t. $g$ and maximize $V(d, g)$ w.r.t. $d$. Update $d$ and $g$ *simultaneously*. Write the update in the form

$$[d_{k+1}, g_{k+1}]^\top = A[d_k, g_k]^\top$$

and specify the $2 \times 2$ matrix $A$.

(b) **Stationary Points** This optimization characterizes a map which has a stationary point [2], what are the coordinates of the stationary points ?

(c) **Extragradient Method** The extragradient method computes an intermediate step :

$$d' = d_k + \alpha g_k, \quad g' = g_k - \alpha d_k$$

and then updates using the gradient evaluated at $(d', g')$ :

$$d_{k+1} = d_k + \alpha g', \quad g_{k+1} = g_k - \alpha d'.$$

Express the extragradient update in matrix form :

$$[d_{k+1}, g_{k+1}]^\top = C[d_k, g_k]^\top,$$

compute the eigenvalues of $C$, and discuss whether the method stabilizes the dynamics.

3.2 **The Motivation for Wasserstein GAN (WGAN) [1]**

The standard GAN objective is given by :

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_r}[\log D(x)] + \mathbb{E}_{z \sim p_z}[\log(1 - D(G(z)))].$$

(a) Show that for a fixed generator $G$, the optimal discriminator is :

$$D^*(x) = \frac{p_r(x)}{p_r(x) + p_g(x)}.$$

(b) Substitute $D^*(x)$ in $V(D, G)$ and show the min-max objective turns to :

$$V(D^*, G) = 2 \cdot \mathrm{JS}(p_r \| p_g) - \log(4).$$

(c) Explain mathematically why gradients vanish when $p_r$ and $p_g$ have disjoint support.

2. A stationary point is a point on the surface of the graph (of the function) where all its partial derivatives are zero (equivalently, the gradient is zero). Source : https ://en.wikipedia.org/wiki/Stationary_pointhttps ://en.wikipedia.org/wiki/Stationary_point

### 3.3 Wasserstein Distance as the Loss Function

WGAN solves the issue discussed in Question 3.2 by defining the objective as the Wasserstein distance between samples from true distribution and the distribution of generated samples.

$$W(p_r, p_g) = \inf_{\gamma \sim \Pi(p_r, p_g)} \mathbb{E}_{(x,y)\sim\gamma}[\|x - y\|].$$

where $x \sim p_r$ and $y \sim p_g$ denote samples from the real and generated distributions, respectively, and $\Pi(p_r, p_g)$ denotes the set of all joint distributions $\gamma(x, y)$ whose marginals are $p_r$ and $p_g$. By using Kantorovich–Rubinstein duality, we can reformulate this distance as :

$$W(p_r, p_g) = \sup_{\|f\|_L \leq 1} \mathbb{E}_{x\sim p_r}[f(x)] - \mathbb{E}_{x\sim p_g}[f(x)].$$

Finally we can write the WGAN objective as :

$$\min_G \max_{D\in\mathcal{D}} \mathbb{E}_{x\sim p_r}[D(x)] - \mathbb{E}_{z\sim p_z}[D(G(z))],$$

where $\mathcal{D}$ is the set of all 1-Lipschitz functions.

(a) Explain mathematically why Wasserstein distance provides meaningful gradients even when supports are disjoint.

(b) In practice, WGAN enforces the 1-Lipschitz constraint using a gradient penalty of the form :

$$\lambda \mathbb{E}_{\hat{x}} \left(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1\right)^2.$$

Here :

- $\hat{x} = \epsilon x + (1 - \epsilon)y$ is a point sampled along the straight line between a real sample $x \sim p_r$ and a generated sample $y \sim p_g$, with $\epsilon \sim \mathcal{U}(0, 1)$,
- $\nabla_{\hat{x}} D(\hat{x})$ denotes the gradient of the critic $D$ with respect to its input,
- $\|\cdot\|_2$ is the Euclidean norm,
- $\lambda > 0$ is a regularization coefficient.

Show that penalizing deviations of $\|\nabla_{\hat{x}} D(\hat{x})\|_2$ from 1 encourages $D$ to be 1-Lipschitz, i.e.,

$$|D(x) - D(y)| \leq \|x - y\| \quad \forall x, y.$$

**Question 4. Policy Gradient Derivation and Variance Reduction**

Let $\pi_\theta(a|x)$ be a stochastic policy with parameters $\theta$. The objective of policy search algorithms is to find a policy which maximizes the following expected discounted return(reward) :

$$J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta(\tau)}\Big[\sum_{t=0}^{T-1}\gamma^t r_t\Big].$$

where expectation is over trajectories $\tau$ of length $T$ with probability $\pi_\theta(\tau)$.

4.1 Show that direct optimization of the $\theta$ parameters of the policy leads to the following gradient formula :

$$\nabla_\theta J(\theta) = \mathbb{E}_\tau\Big[G\sum_{t=0}^{T-1}\nabla_\theta \log \pi_\theta(a_t|x_t)\Big], \quad G = \sum_{t=0}^{T-1}\gamma^t r_t.$$

4.2 Explain why this gradient estimator has high variance. (Hint : For each time step $t$, divide the formula into two parts : one considering the return from the beginning up to time $t$, and the other considering the return from time $t$ to the end. Then, analyze the variance based on variance/covariance of these terms.)

Then Derive an equivalent expression using the standard *causal return* definition :

$$G_t = \sum_{t'=t}^{T}\gamma^{t'-t}\, r_{t'}.$$

(Hint : Analyze the gradient derived in Question 4.1 and determine when the policy gradient increases or decreases the probability of taking action $a_t$ at state $x_t$. Then reflect on how the different terms in the return $G$ influence this update at time $t$, and argue why the causal form is more appropriate.)

4.3 Although causal return calculation reduces the variance, it does not lead to the minimum variance possible. To achieve this goal, we subtract a state dependent term called baseline. The gradient estimator will have the following form :

$$\hat{g} = \sum_t (G_t - b(x_t))\nabla_\theta \log \pi_\theta(a_t|x_t)$$

Show that this estimator is unbiased :

$$\nabla_\theta J(\theta) = \mathbb{E}_\tau\Big[\hat{g}\Big]$$

4.4 Derive the optimal baseline $b^*(x_t)$ in terms of $\nabla_\theta \log \pi_\theta(a_t \mid x_t)$ and $G_t$ and justify mathematically that this optimum baseline leads to the minimum variance.

4.5 Could the baseline be state-action dependent ?

**Question 5. Sharpening and Diversity in KL-regularized RLVR**

Consider a single prompt $x$ and a finite set of completions

$$\{y_1, \ldots, y_n\}. \tag{6}$$

Let a verifiable reward $R(x, y_i) \in \{0, 1\}$ be given for each completion $y_i$. We consider distributions $q(\cdot \mid x)$ over the $y_i$ and a fixed reference distribution $p_{\text{ref}}(\cdot \mid x)$. For brevity, write

$$q_i = q(y_i \mid x), \quad p_{\text{ref},i} = p_{\text{ref}}(y_i \mid x), \quad R_i = R(x, y_i). \tag{7}$$

We define the KL-penalized RLVR objective :

$$J(q) = \sum_{i=1}^{n} q_i R_i - \beta \sum_{i=1}^{n} q_i \log \frac{q_i}{p_{\text{ref},i}}, \quad \beta > 0 \tag{8}$$

subject to $q_i \geq 0$ and $\sum_{i=1}^{n} q_i = 1$.

5.1 Derive the optimal distribution $q^*$ by solving the constrained maximization problem for $J(q)$, and show that

$$q_i^* \propto p_{\text{ref},i} \exp\left(\frac{R_i}{\beta}\right). \tag{9}$$

5.2 Assume that at least one completion has reward 1 and at least one has reward 0. Prove that as $\beta \to 0$, the distribution $q^*$ concentrates on the set of maximally rewarded completions

$$\mathcal{I}_1 = \{i : R_i = 1\}. \tag{10}$$

in the sense that for every $j$ with $R_j = 0$, we have $q_j^* \to 0$, and the total mass on $\mathcal{I}_1$ tends to 1.

5.3 Using the explicit formula for $q^*$, for unrewarded completions derive a quantitative lower bound of the form

$$q_j^* \geq \epsilon \quad \Rightarrow \quad \beta \geq \beta_{\min}(\epsilon, p_{\text{ref}}, R) \tag{11}$$

for some expression $\beta_{\min}(\cdot)$ that depends on the ratio

$$\frac{p_{\text{ref},j}}{\sum_{k \in \mathcal{I}_1} p_{\text{ref},k}}. \tag{12}$$

5.4 Interpret the result as a theoretical statement about *diversity collapse* and *distributional sharpening* in RLVR with strict verifiable rewards, and discuss qualitatively how larger $\beta$ (or additional entropy regularization) can effect that.

# Références

[1] Martin Arjovsky, Soumith Chintala, and Léon Bottou. Wasserstein generative adversarial networks. In *International conference on machine learning*, pages 214–223. Pmlr, 2017.